Input data file for analysis and design using STAAD Pro is given here below:

```
STAAD SPACE taxiSh
STAAD SPACE LINKWAY 201911
START JOB INFORMATION
ENGINEER DATE 26-Nov-19
END JOB INFORMATION
INPUT WIDTH 79
PAGE LENGTH 15000
SET NL 1000000
UNIT METER KN
JOINT COORDINATES
1 0 0 0; 2 4.5 0 0; 3 9 0 0; 4 13.5 0 0; 5 0 3 -0.5; 6 0 3 0; 7 0 3 0.5;
8 0 3 1; 9 0 3 1.5; 10 0 3 2; 11 0 3 2.5; 12 0 3 3.1; 13 1.5 3 -0.5;
14 1.5 3 1.05964e-008; 15 1.5 3 3.1; 16 3 3 -0.5; 17 3 3 1.05964e-008;
18 3 3 3.1; 19 4.5 3 -0.5; 20 4.5 3 0; 21 4.5 3 0.5; 22 4.5 3 1; 23 4.5 3 1.5;
24 4.5 3 2; 25 4.5 3 2.5; 26 4.5 3 3.1; 27 6 3 -0.5; 28 6 3 1.05964e-008;
29 6 3 3.1; 30 7.5 3 -0.5; 31 7.5 3 1.05964e-008; 32 7.5 3 3.1; 33 9 3 -0.5;
34 9 3 0; 35 9 3 0.5; 36 9 3 1; 37 9 3 1.5; 38 9 3 2; 39 9 3 2.5; 40 9 3 3.1;
41 10.5 3 -0.5; 42 10.5 3 1.05964e-008; 43 10.5 3 3.1; 44 12 3 -0.5;
45 12 3 1.05964e-008; 46 12 3 3.1; 47 13.5 3 -0.5; 48 13.5 3 0; 49 13.5 3 0.5;
50 13.5 3 1; 51 13.5 3 1.5; 52 13.5 3 2; 53 13.5 3 2.5; 54 13.5 3 3.1;
MEMBER INCIDENCES
1 1 6; 2 2 20; 3 3 34; 4 4 48; 5 5 6; 6 6 7; 7 7 8; 8 8 9; 9 9 10; 10 10 11;
11 11 12; 12 13 14; 13 14 15; 14 16 17; 15 17 18; 16 19 20; 17 20 21; 18 21 22;
19 22 23; 20 23 24; 21 24 25; 22 25 26; 23 27 28; 24 28 29; 25 30 31; 26 31 32;
27 33 34; 28 34 35; 29 35 36; 30 36 37; 31 37 38; 32 38 39; 33 39 40; 34 41 42;
35 42 43; 36 44 45; 37 45 46; 38 47 48; 39 48 49; 40 49 50; 41 50 51; 42 51 52;
43 52 53; 44 53 54; 45 5 13; 46 13 16; 47 16 19; 48 19 27; 49 27 30; 50 30 33;
51 33 41; 52 41 44; 53 44 47; 54 6 14; 55 17 20; 56 20 28; 57 31 34; 58 34 42;
59 45 48; 60 14 17; 61 28 31; 62 42 45; 63 12 15; 64 15 18; 65 18 26; 66 26 29;
67 29 32; 68 32 40; 69 40 43; 70 43 46; 71 46 54;
START GROUP DEFINITION
MEMBER
_COL 1 TO 4
_MAINC 5 TO 11 16 TO 22 27 TO 33 38 TO 44
_TIE 54 TO 62
_EAVES 45 TO 53 63 TO 71
_IRAFTER 12 TO 15 23 TO 26 34 TO 37
END GROUP DEFINITION
SUPPORTS
1 TO 4 FIXED
START USER TABLE
TABLE 1
UNIT METER KN
* inverted U - cut from RHS250x150x12
CHANNEL
C_1
0.003912 0.15 0.012 0.1 0.012 1.34556e-005 3.81399e-006 1.87776e-007 -
0.0329939 0.0018 0.0016
C_2
0.004392 0.15 0.012 0.12 0.012 1.57466e-005 6.36529e-006 2.10816e-007 -
0.0414098 0.0018 0.00192
C_3
0.004872 0.15 0.012 0.14 0.012 1.80377e-005 9.77728e-006 2.33856e-007 -
0.0501379 0.0018 0.00224
C_4
0.005352 0.15 0.012 0.16 0.012 2.03287e-005 1.41507e-005 2.56896e-007 -
0.0590942 0.0018 0.00256
C_5
0.005832 0.15 0.012 0.18 0.012 2.26197e-005 1.95849e-005 2.79936e-007 -
0.0682222 0.0018 0.00288
C_6
0.006312 0.15 0.012 0.2 0.012 2.49108e-005 2.61779e-005 3.02976e-007 -
0.0774829 0.0018 0.0032
END
DEFINE MATERIAL START
ISOTROPIC S355
```

```
E 2.05e+008
POISSON 0.3
DENSITY 77
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH FY 35500 FU 510000 RY 1.5 RT 1.2
ISOTROPIC S275
E 2.05e+008
POISSON 0.3
DENSITY 77
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH FY 27500 FU 430000 RY 1.5 RT 1.2
ISOTROPIC M40
E 3.33e+007
POISSON 0.2
DENSITY 25
ALPHA 1.2e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 40000
END DEFINE MATERIAL
MEMBER PROPERTY BRITISH
45 TO 53 63 TO 71 TABLE ST UA100X100X15
12 TO 15 23 TO 26 34 TO 37 54 TO 62 TABLE ST TUB80805.0
11 22 33 44 UPTABLE 1 C_1
10 21 32 43 UPTABLE 1 C_2
9 20 31 42 UPTABLE 1 C_3
8 19 30 41 UPTABLE 1 C_4
5 16 27 38 UPTABLE 1 C_5
6 7 17 18 28 29 39 40 UPTABLE 1 C_6
MEMBER PROPERTY BRITISH
1 TO 4 TABLE ST TUB25015012.0
CONSTANTS
*BETA 45 MEMB 45 TO 53 63 TO 71
*BETA 90 MEMB 1 TO 11 16 TO 22 27 TO 33 38 TO 44
MATERIAL S355 ALL
LOAD 1 LOADTYPE Dead  TITLE DL
SELFWEIGHT Y -1
floor load
yrange 2.99 3.01 fload -1
LOAD 2 LOADTYPE Live  TITLE LL
floor load
yrange 2.99 3.01 fload -.5
LOAD 3 LOADTYPE Wind  TITLE WD
floor load
yrange 2.99 3.01 fload -.9
LOAD 4 LOADTYPE Wind  TITLE WU
floor load
yrange 2.99 3.01 fload .9
LOAD COMB 31 SLS1
1 1.0 2 1.0
LOAD COMB 32 SLS2
1 1.0 3 1.0
LOAD COMB 33 SLS3
1 1.0 4 1.0
LOAD COMB 51 ULS1
1 1.35 2 1.5
LOAD COMB 52 ULS2
1 1.35 2 1.5 3 0.75
LOAD COMB 53 ULS3
1 1.0 4 1.5
PERFORM ANALYSIS PRINT STATICS CHECK
LOAD LIST 31 TO 33
PRINT JOINT DISPLACEMENTS
PRINT SUPPORT REACTION ALL
```

```
LOAD LIST 51 TO 53
PRINT MEMBER FORCES ALL
PARAMETER 1
CODE EN 1993-1-1:2005
* NA 0:orig, 1: BS, 4:NF, 7:SS, 9:MS
NA 7 all
* sgr 3: S235, 1: S275, 2: S355, 3: S420, 4: S460
SGR 2 all
*ly 4.4 member 64 to 67
*lz 4.4 member 64 to 67
*cmm 4 member 17 18
TRACK 2 all
fixed group
group mem 1 TO 4
group mem 5 TO 11 16 TO 22 27 TO 33 38 TO 44
group mem 54 TO 62
group mem 45 TO 53 63 TO 71
group mem 12 TO 15 23 TO 26 34 TO 37
CHECK CODE all
SELECT OPTIMIZED
PERFORM ANALYSIS PRINT STATICS CHECK
CHECK CODE all
PRINT ANALYSIS RESULTS
STEEL TAKE OFF
FINISH
```

Results of the analysis and design using STAAD Pro are given here below: